DATA*6200 F25 Assignment2

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1 Introduction

In this assignment, we analyze British Columbia wildfire incident data from 2012 to 2024. Our primary dataset is derived from KMZ files published by the BC Wildfire Service, and we supplement it with wildfire summary tables scraped from the BC Wildfire website for comparison and validation. After cleaning, consolidating, and parsing the KMZ records, we investigate major wildfire trends across years and explore the relationship between wildfire characteristics and causes.

2 Preparation

Before working with the data, we first install and load the necessary R packages for this project:

```
library(sf)
library(tidyverse)
library(httr)
library(xml2)
library(rvest)
library(tmap)
library(leaflet)
library(bcmaps)  # This has been approved by professor
library(patchwork)
```

3 Get and Inspect BC Wildfire Data (Files)

3.1 Download & Read KMZ Files

The files used in this assignment are available at the following resource: [the wildfire KMZ datasets published by the British Columbia government](). We include both the main and archived files, since the primary set does not cover all target years, particularly 2012 and 2023. The specific files we use are listed below:

```
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fires/Archive/BC%20Fire%20Points%202010-2012.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fires/Archive/BC%20Fire%20Points%202011-2013.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fires/Archive/BC%20Fire%20Points%202012-2014.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
```

```
s/Archive/BC%20Fire%20Points%202013-2015.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/Archive/BC%20Fire%20Points%202014-2016.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/Archive/BC%20Fire%20Points%202016-2018.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/Archive/BC%20Fire%20Points%202023.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202013-2015.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202015-2017.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202017-2019.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202018-2020.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202019-2021.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202020-2022.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202021-2023.kmz
https://www.for.gov.bc.ca/ftp/HPR/external/!publish/Maps_and_Data/GoogleEarth/WMB_Fire
s/BC%20Fire%20Points%202024.kmz
```

We read all files as sf objects. Because each KMZ file contains multiple layers by year, an sf object is created for each year layer:

▶ Code

```
kmz/BC_Fire_Points_2010-2012/2012
kmz/BC_Fire_Points_2010-2012/2011
kmz/BC_Fire_Points_2010-2012/2010
kmz/BC_Fire_Points_2011-2013/2013
kmz/BC_Fire_Points_2011-2013/2012
kmz/BC_Fire_Points_2011-2013/2011
kmz/BC_Fire_Points_2012-2014/2014
kmz/BC_Fire_Points_2012-2014/2013
kmz/BC_Fire_Points_2012-2014/2012
kmz/BC_Fire_Points_2013-2015/2015
kmz/BC_Fire_Points_2013-2015/2014
kmz/BC_Fire_Points_2013-2015/2013
kmz/BC_Fire_Points_2014-2016/2014
kmz/BC_Fire_Points_2014-2016/2015
kmz/BC_Fire_Points_2014-2016/2016
kmz/BC_Fire_Points_2016-2018/2018
kmz/BC_Fire_Points_2016-2018/2017
kmz/BC_Fire_Points_2016-2018/2016
kmz/BC_Fire_Points_2023/2023
kmz/BC_Fire_Points_2015-2017/2015
kmz/BC_Fire_Points_2015-2017/2016
kmz/BC_Fire_Points_2015-2017/2017
kmz/BC_Fire_Points_2017-2019/2019
kmz/BC_Fire_Points_2017-2019/2018
kmz/BC_Fire_Points_2017-2019/2017
```

```
kmz/BC_Fire_Points_2018-2020/2020
kmz/BC_Fire_Points_2018-2020/2019
kmz/BC_Fire_Points_2018-2020/2018
kmz/BC_Fire_Points_2019-2021/2021
kmz/BC_Fire_Points_2019-2021/2020
kmz/BC_Fire_Points_2019-2021/2019
kmz/BC_Fire_Points_2020-2022/2022
kmz/BC_Fire_Points_2020-2022/2021
kmz/BC_Fire_Points_2020-2022/2020
kmz/BC_Fire_Points_2021-2023/2023
kmz/BC_Fire_Points_2021-2023/2022
kmz/BC_Fire_Points_2021-2023/2021
kmz/BC_Fire_Points_2024/2024
```

3.2 Inspect Description

Each sf object contains three columns: **Name**, **Description**, and **geometry**. Most of the meaningful information is stored inside the **Description** field. (Shown below as a dataframe for interpretability.)

	kmz.BC_Fire_Points_2010.2012.2012.Name <chr>
1	2012-V10011

1 row | 1-2 of 4 columns

The Description content has an XML structure. It includes key information such as **year**, **fire number**, **latitude**, **longitude**, **fire size** and **cause**, and may include additional attributes depending on the year. Here, we check whether the year in each Description matches the year of its file layer. We count how many Description entries contain the expected year relative to the total:

▶ Code

```
1660/1660  kmz/BC_Fire_Points_2010-2012/2012
 655/ 655  kmz/BC_Fire_Points_2010-2012/2011
1673/1673  kmz/BC_Fire_Points_2010-2012/2010
1852/1852  kmz/BC_Fire_Points_2011-2013/2013
1650/1650  kmz/BC_Fire_Points_2011-2013/2012
 653/ 653  kmz/BC_Fire_Points_2011-2013/2011
1465/1465  kmz/BC_Fire_Points_2012-2014/2014
1861/1861  kmz/BC_Fire_Points_2012-2014/2013
1649/1649  kmz/BC_Fire_Points_2012-2014/2012
1859/1859  kmz/BC_Fire_Points_2013-2015/2015
1465/1465  kmz/BC_Fire_Points_2013-2015/2014
1861/1861  kmz/BC_Fire_Points_2013-2015/2013
1465/1465  kmz/BC_Fire_Points_2014-2016/2014
1858/1858  kmz/BC_Fire_Points_2014-2016/2015
3147/3147  kmz/BC_Fire_Points_2014-2016/2016
2115/2115  kmz/BC_Fire_Points_2016-2018/2018
2344/2344  kmz/BC_Fire_Points_2016-2018/2017
```

```
2121/2121  kmz/BC_Fire_Points_2016-2018/2016
   0/2293  kmz/BC_Fire_Points_2023/2023
1858/1858  kmz/BC_Fire_Points_2015-2017/2015
1057/1057  kmz/BC_Fire_Points_2015-2017/2016
1353/1353  kmz/BC_Fire_Points_2015-2017/2017
1871/1871  kmz/BC_Fire_Points_2017-2019/2019
3209/3209  kmz/BC_Fire_Points_2017-2019/2018
2344/2344  kmz/BC_Fire_Points_2017-2019/2017
   0/ 670  kmz/BC_Fire_Points_2018-2020/2020
   0/ 825  kmz/BC_Fire_Points_2018-2020/2019
   0/2110  kmz/BC_Fire_Points_2018-2020/2018
   0/1648  kmz/BC_Fire_Points_2019-2021/2021
   0/ 670  kmz/BC_Fire_Points_2019-2021/2020
   0/ 825  kmz/BC_Fire_Points_2019-2021/2019
1802/1802  kmz/BC_Fire_Points_2020-2022/2022
1647/1647  kmz/BC_Fire_Points_2020-2022/2021
 670/ 670  kmz/BC_Fire_Points_2020-2022/2020
   0/2296  kmz/BC_Fire_Points_2021-2023/2023
   0/1803  kmz/BC_Fire_Points_2021-2023/2022
   0/1647  kmz/BC_Fire_Points_2021-2023/2021
1698/1698  kmz/BC_Fire_Points_2024/2024
```

The result reveals an all-or-nothing situation. For example, in *kmz/BC_Fire_Points_2023/2023*, every Description entry is empty. Therefore, we need to inspect Description emptiness by KMZ year layer:

	kmz.BC_Fire_Points_2023.2023.Name <chr>
1	C23369

1 row | 1-2 of 4 columns

3.3 Find Empty Descripton

We now count empty Description entries per KMZ year layer:

▶ Code

```
1660/1660  kmz/BC_Fire_Points_2010-2012/2012
 655/ 655  kmz/BC_Fire_Points_2010-2012/2011
1673/1673  kmz/BC_Fire_Points_2010-2012/2010
1852/1852  kmz/BC_Fire_Points_2011-2013/2013
1650/1650  kmz/BC_Fire_Points_2011-2013/2012
 653/ 653  kmz/BC_Fire_Points_2011-2013/2011
1465/1465  kmz/BC_Fire_Points_2012-2014/2014
1861/1861  kmz/BC_Fire_Points_2012-2014/2013
1649/1649  kmz/BC_Fire_Points_2012-2014/2012
1859/1859  kmz/BC_Fire_Points_2013-2015/2015
1465/1465  kmz/BC_Fire_Points_2013-2015/2014
1861/1861  kmz/BC_Fire_Points_2013-2015/2013
```

```
1465/1465  kmz/BC_Fire_Points_2014-2016/2014
1858/1858  kmz/BC_Fire_Points_2014-2016/2015
3147/3147  kmz/BC_Fire_Points_2014-2016/2016
2115/2115  kmz/BC_Fire_Points_2016-2018/2018
2344/2344  kmz/BC_Fire_Points_2016-2018/2017
2121/2121  kmz/BC_Fire_Points_2016-2018/2016
   0/2293  kmz/BC_Fire_Points_2023/2023
1858/1858  kmz/BC_Fire_Points_2015-2017/2015
1057/1057  kmz/BC_Fire_Points_2015-2017/2016
1353/1353  kmz/BC_Fire_Points_2015-2017/2017
1871/1871  kmz/BC_Fire_Points_2017-2019/2019
3209/3209  kmz/BC_Fire_Points_2017-2019/2018
2344/2344  kmz/BC_Fire_Points_2017-2019/2017
   0/ 670  kmz/BC_Fire_Points_2018-2020/2020
   0/ 825  kmz/BC_Fire_Points_2018-2020/2019
   0/2110  kmz/BC_Fire_Points_2018-2020/2018
   0/1648  kmz/BC_Fire_Points_2019-2021/2021
   0/ 670  kmz/BC_Fire_Points_2019-2021/2020
   0/ 825  kmz/BC_Fire_Points_2019-2021/2019
1802/1802  kmz/BC_Fire_Points_2020-2022/2022
1647/1647  kmz/BC_Fire_Points_2020-2022/2021
 670/ 670  kmz/BC_Fire_Points_2020-2022/2020
   0/2296  kmz/BC_Fire_Points_2021-2023/2023
   0/1803  kmz/BC_Fire_Points_2021-2023/2022
   0/1647  kmz/BC_Fire_Points_2021-2023/2021
1698/1698  kmz/BC_Fire_Points_2024/2024
```

The result mirrors the year-matching result: if a layer is missing one Description value, it is missing all of them. Because these layers provide almost no usable information, they should be excluded from our analysis.

3.4 Observe the same year datum from different files.

Before excluding layers with empty Description, we compare duplicate KMZ layers representing the same year. For example, both *kmz/BC_Fire_Points_2013-2015/2015* and *kmz/BC_Fire_Points_2015-2017/2015* contain wildfire data for 2015, but with different record counts. Let’s examine the difference:

▶ Code

	Name <chr>
513	G80528

1 row | 1-2 of 5 columns

The difference consists of only one record, matching the record count discrepancy. This record was removed in *kmz/BC_Fire_Points_2015-2017/2015*, but we cannot determine a specific reason. In general, multiple versions of the same year differ only slightly.

3.5 Select Target Data

Given multiple KMZ layers for some years, we now choose which ones to use. Ideally, we would rely on the non-archived (main) dataset. However, 2012 and 2023 are missing from the main page, so archived files must be used. After reviewing, 2012, 2015, and 2023 still have multiple KMZ year layers. Since differences between these datasets are small, we choose the newest version under the assumption of updates. Even so, 2023 still has empty Description values.

Final selection rules:

1. Prefer non-archived files.
2. Exclude layers with empty Description.
3. Prefer the newest version of a year.
4. For years not available otherwise, use archived or empty-Description files.

Archive?	Is target?	Empty Description?	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	delete?	non-empty-description-rows/total-rows	KMZ year layer
yes			x														1660/1660	kmz/BC_Fire_Points_2010-2012/2012
yes	no															delete	655/ 655	kmz/BC_Fire_Points_2010-2012/2011
yes	no															delete	1673/1673	kmz/BC_Fire_Points_2010-2012/2010
yes				x													1852/1852	kmz/BC_Fire_Points_2011-2013/2013
yes			x														1650/1650	kmz/BC_Fire_Points_2011-2013/2012
yes	no															delete	653/ 653	kmz/BC_Fire_Points_2011-2013/2011
yes					x												1465/1465	kmz/BC_Fire_Points_2012-2014/2014
yes				x													1861/1861	kmz/BC_Fire_Points_2012-2014/2013
yes			x														1649/1649	kmz/BC_Fire_Points_2012-2014/2012
						x											1859/1859	kmz/BC_Fire_Points_2013-2015/2015
					x												1465/1465	kmz/BC_Fire_Points_2013-2015/2014
				x													1861/1861	kmz/BC_Fire_Points_2013-2015/2013
yes					x												1465/1465	kmz/BC_Fire_Points_2014-2016/2014
yes						x											1858/1858	kmz/BC_Fire_Points_2014-2016/2015
yes							x										3147/3147	kmz/BC_Fire_Points_2014-2016/2016
yes									x								2115/2115	kmz/BC_Fire_Points_2016-2018/2018
yes								x									2344/2344	kmz/BC_Fire_Points_2016-2018/2017
yes							x										2121/2121	kmz/BC_Fire_Points_2016-2018/2016
yes		yes												x		delete	0/2293	kmz/BC_Fire_Points_2023/2023
						x											1858/1858	kmz/BC_Fire_Points_2015-2017/2015
							x										1057/1057	kmz/BC_Fire_Points_2015-2017/2016
								x								delete	1353/1353	kmz/BC_Fire_Points_2015-2017/2017
										x							1871/1871	kmz/BC_Fire_Points_2017-2019/2019
									x								3209/3209	kmz/BC_Fire_Points_2017-2019/2018
								x									2344/2344	kmz/BC_Fire_Points_2017-2019/2017
		yes									x					delete	0/ 670	kmz/BC_Fire_Points_2018-2020/2020
		yes								x						delete	0/ 825	kmz/BC_Fire_Points_2018-2020/2019
		yes							x							delete	0/2110	kmz/BC_Fire_Points_2018-2020/2018
		yes										x				delete	0/1648	kmz/BC_Fire_Points_2019-2021/2021
		yes									x					delete	0/ 670	kmz/BC_Fire_Points_2019-2021/2020
		yes								x						delete	0/ 825	kmz/BC_Fire_Points_2019-2021/2019
													x				1802/1802	kmz/BC_Fire_Points_2020-2022/2022
												x					1647/1647	kmz/BC_Fire_Points_2020-2022/2021
											x						670/ 670	kmz/BC_Fire_Points_2020-2022/2020
		yes												x			0/2296	kmz/BC_Fire_Points_2021-2023/2023
		yes											x			delete	0/1803	kmz/BC_Fire_Points_2021-2023/2022
		yes										x				delete	0/1647	kmz/BC_Fire_Points_2021-2023/2021
															x		1698/1698	kmz/BC_Fire_Points_2024/2024

Target KMZ year layers

▶ Code

```
kmz/BC_Fire_Points_2012-2014/2012
kmz/BC_Fire_Points_2013-2015/2013
kmz/BC_Fire_Points_2013-2015/2014
kmz/BC_Fire_Points_2015-2017/2015
kmz/BC_Fire_Points_2015-2017/2016
```

```
kmz/BC_Fire_Points_2017-2019/2017
kmz/BC_Fire_Points_2017-2019/2018
kmz/BC_Fire_Points_2017-2019/2019
kmz/BC_Fire_Points_2020-2022/2020
kmz/BC_Fire_Points_2020-2022/2021
kmz/BC_Fire_Points_2020-2022/2022
kmz/BC_Fire_Points_2021-2023/2023
kmz/BC_Fire_Points_2024/2024
```

3.6 Check Description Format

Now that we have selected one dataset per year, we inspect the structure of Description to extract required properties. Each year uses slightly different tag names. We assume all records within a given year share the same XML structure, so we inspect one sample record per year:

▶ Code

```
[1] "FIREYEAR: 2012"
[1] "FIRENUMBER: V10011"
[1] "FIRETYPE: Fire"
[1] "LATITUDE: 49.381218"
[1] "LONGITUDE: -121.553635"
[1] "CURRENTSIZE: 0.009"
[1] "CAUSE_GENERAL_KMZ: Human"

[1] "FIREYEAR: 2013"
[1] "FIRENUMBER: V10012"
[1] "FIRETYPE: Fire"
[1] "LATITUDE: 49.237083"
[1] "LONGITUDE: -122.002815"
[1] "CURRENTSIZE: 0.2"
[1] "CAUSE_GENERAL_KMZ: Human"

[1] "FIREYEAR: 2014"
[1] "FIRENUMBER: V10010"
[1] "FIRETYPE: Fire"
[1] "LATITUDE: 49.377018"
[1] "LONGITUDE: -123.142685"
[1] "CURRENTSIZE: 0.009"
[1] "CAUSE_GENERAL_KMZ: Lightning"

[1] "FIREYEAR: 2015"
[1] "FIRENUMBER: V10146"
[1] "FIRETYPE: Fire"
[1] "LATITUDE: 50.003601"
[1] "LONGITUDE: -121.516083"
[1] "CURRENTSIZE: 0.1"
[1] "CAUSE GENERAL: Person"

[1] "FIREYEAR: 2016"
[1] "FIRENUMBER: V10024"
```

```
[1] "FIRETYPE: Fire"
[1] "LATITUDE: 49.376701"
[1] "LONGITUDE: -121.84005"
[1] "CURRENTSIZE: 4.3"
[1] "CAUSE GENERAL: Person"

[1] "FIRE_NUMBER: N11984"
[1] "FIRE_YEAR: 2017"
[1] "IGNITION_DATE: 2017/08/31"
[1] "GEOGRAPHIC_DESCRIPTION: Elder Creek"
[1] "LATITUDE: 49.0044"
[1] "LONGITUDE: -114.4116"
[1] "CURRENT_SIZE: 903.1"

[1] "FIRE_NUMBER: N11605"
[1] "FIRE_YEAR: 2018"
[1] "IGNITION_DATE: 2018/07/29"
[1] "GEOGRAPHIC_DESCRIPTION: Flathead"
[1] "LATITUDE: 49.0186"
[1] "LONGITUDE: -114.4319"
[1] "CURRENT_SIZE: 0.009"

[1] "FIRE_NUMBER: N10562"
[1] "FIRE_YEAR: 2019"
[1] "IGNITION_DATE: 2019/06/13"
[1] "GEOGRAPHIC_DESCRIPTION: Red Canyon Creek"
[1] "LATITUDE: 49.1418"
[1] "LONGITUDE: -115.088"
[1] "CURRENT_SIZE: 0.009"

[1] "YEAR: 2020"
[1] "Fire Number: N50404"
[1] "IGNITION DATE: 2020-06-22 6:15:00 AM"
[1] "GEOGRAPHIC: Marsh Creek Road"
[1] "LATITUDE: 49.146599"
[1] "LONGITUDE: -117.522514"
[1] "FIRE_SIZE_HA: 0.009"
[1] "INCIDENT_NAME: <Null>"
[1] "FIRE_OF_NOTE_IND: N"
[1] "RESPONSE_TYPE_DESC: Full"

[1] "YEAR: 2021"
[1] "Fire Number: C50366"
[1] "IGNITION DATE: 2021-04-21 1:30:49 PM"
[1] "GEOGRAPHIC: Elkins Ranch"
[1] "LATITUDE: 51.48205"
[1] "LONGITUDE: -123.824717"
[1] "FIRE_SIZE_HA: 38.5"
[1] "INCIDENT_NAME: <Null>"
[1] "FIRE_OF_NOTE_IND: N"
[1] "RESPONSE_TYPE_DESC: Full"

[1] "YEAR: 2022"
[1] "Fire Number: C31003"
```

```
[1] "IGNITION DATE: 2022-07-17 5:08:40 PM"
[1] "GEOGRAPHIC: South of Amos Creek"
[1] "LATITUDE: 52.623"
[1] "LONGITUDE: -121.019467"
[1] "FIRE_SIZE_HA: 0.09"
[1] "INCIDENT_NAME: <Null>"
[1] "FIRE_OF_NOTE_IND: N"
[1] "RESPONSE_TYPE_DESC: Modified"
```

```
empty description (2023)
```

```
[1] "YEAR: 2024"
[1] "Fire Number: C10084"
[1] "IGNITION DATE: 2024-04-17 10:36:45 AM"
[1] "GEOGRAPHIC: NW of Kluskus Lakes"
[1] "LATITUDE: 53.087433"
[1] "LONGITUDE: -124.49085"
[1] "FIRE_SIZE_HA: 2.28"
[1] "INCIDENT_NAME: C10084"
[1] "FIRE_OF_NOTE_IND: N"
[1] "RESPONSE_TYPE_DESC: Full"
[1] "WAS_FIRE_OF_NOTE_IND: N"
```

The available properties differ across years, more recent years contain more details. For this assignment, we only need **year**, **fire number**, **latitude**, **longitude**, **fire size** and **cause**. Because tag names vary, we standardize them as follows:

- “FIREYEAR”, “YEAR” -> “FIRE_YEAR”
- “FIRENUBER”, “Fire Number” -> “FIRE NUMBER”
- “CURRENT SIZE”, “FIRE_SIZE_HA” -> “CURRENT_SIZE”
- “CAUSE_GENERAL_KMZ”, “CAUSE GENERAL” -> “CAUSE_GENERAL”

Before unifying names, we first bind all KMZ year-layer sf objects into a single object.

3.7 Bind sf Objects

Here, we combine all the KMZ year layers into one sf object:

```
wfbc_target_sf <- bind_rows(unname(wfbc_target_sf_nlist))
```

```
the number of rows: 23427
```

3.8 Parse Description

We now parse Description to extract and clean wildfire information. First, we apply the unified property names. Then, we convert data types appropriately for analysis:

▶ Code

	year	number	latitude	longitude	size_ha	cause	geometry
	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<sf_POINT>
1	2012	V10011	49.38122	-121.5536	0.009	Human	<sf_POINT>

1 row

3.9 Check Duplicate

Before performing further analysis, we check for duplicate records within the combined sf object:

▶ Code

```
The number of duplicate rows: 0
```

3.10 Check NULL

We also evaluate missing values. First, we check whether **year**, **number**, **latitude**, **longitude** or **geometry** contain any NULL entries:

▶ Code

year	count	geometry
<dbl>	<int>	<s_MULTIP>
2023	2296	<s_MULTIP>

1 row

No records have missing values in **year**, **number**, **latitude**, **longitude** and **geometry**, except for 2023, which contains only geometry information.

Next, we check missing values in **cause**:

▶ Code

year	count	geometry
<dbl>	<int>	<s_MULTIP>
2017	2344	<s_MULTIP>
2018	3209	<s_MULTIP>
2019	1871	<s_MULTIP>
2020	670	<s_MULTIP>

year	count	geometry
<dbl>	<int>	<s_MULTIP>
2021	1647	<s_MULTIP>
2022	1802	<s_MULTIP>
2023	2296	<s_MULTIP>
2024	1698	<s_MULTIP>

8 rows

Although cause is fully recorded from 2012 to 2016, it is missing for all records from 2017 onward.

Lastly, we inspect NULLs in **fire size**:

▶ Code

year	count	geometry
<dbl>	<int>	<s_MULTIP>
2017	6	<s_MULTIP>
2018	17	<s_MULTIP>
2019	179	<s_MULTIP>
2023	2296	<s_MULTIP>

4 rows

There are some missing fire size values between 2017 and 2019, but the number is small. We can exclude them later when analyzing fire size distributions if necessary.

At this point, we have produced a cleaned British Columbia wildfire dataset from the KMZ files. In the next chapter, we will scrape the table data available on the BC Wildfire website to compare it with the dataset we obtained.

4 Scrape BC Wildfire Data on WEB

4.1 Scrape Data

The same BC Wildfire website that hosts the KMZ files also provides a table of major wildfire incidents (over 5 hectares) from 2012 to 2017. We want to compare this table with the KMZ-based dataset. First, we scrape the table data.

```
wfbc_html <- read_html(
  "https://www2.gov.bc.ca/gov/content/safety/wildfire-status/about-bcws/wildfire-stati
  )
```

▶ Code

Year	Fire Number	Fire Centre	Latitude	Longitude
<int>	<chr>	<chr>	<chr>	<chr>
2012	C10075 (2012)	Cariboo	52 44.747	124 22.619
2012	C10118 (2012)	Cariboo	52 41.849	123 00.607
2012	C10247 (2012)	Cariboo	53 03.159	122 58.020
2012	C10250 (2012)	Cariboo	52 30.773	122 26.629
2012	C20005 (2012)	Cariboo	52 07.676	121 54.904
2012	C20009 (2012)	Cariboo	52 06.000	121 59.000
2012	C20016 (2012)	Cariboo	52 06.430	121 59.783
2012	C20047 (2012)	Cariboo	51 54.970	122 40.119
2012	C20104 (2012)	Cariboo	51 49.557	121 51.749
2012	C20108 (2012)	Cariboo	52 09.063	122 18.796

1-10 of 1,108 rows | 1-5 of 8 columns Previous **1** 2 3 4 5 6 ... 111 Next

4.2 Modify Columns

The scraped table contains two issues that need correction before comparison. First, values in the **Fire Number** column include both numbers and embedded year information, so we must extract the numeric fire number and verify that the year inside the number matches the Year column. Second, **Latitude** and **Longitude** are stored as character strings and must be converted to numeric format.

▶ Code

year	number_with_year	fire_center	latitude	longitude
<int>	<chr>	<chr>	<dbl>	<dbl>
2012	C10075 (2012)	Cariboo	52.44747	-124.2262
2012	C10118 (2012)	Cariboo	52.41849	-123.0061
2012	C10247 (2012)	Cariboo	53.03159	-122.5802
2012	C10250 (2012)	Cariboo	52.30773	-122.2663
2012	C20005 (2012)	Cariboo	52.07676	-121.5490
2012	C20009 (2012)	Cariboo	52.06000	-121.5900
2012	C20016 (2012)	Cariboo	52.06430	-121.5978
2012	C20047 (2012)	Cariboo	51.54970	-122.4012
2012	C20104 (2012)	Cariboo	51.49557	-121.5175
2012	C20108 (2012)	Cariboo	52.09063	-122.1880

1-10 of 1,108 rows | 1-5 of 10 columns Previous **1** 2 3 4 5 6 ... 111 Next

Now that we have modified the problems, we want to check whether there is missing latitude or longitude and whether year2 (extracted from Fire Number) corresponds to year:

```
wfbc_1217_tb |>
  filter(
    is.na(latitude) |
    is.na(longitude) |
```

```
    year != year2
  )
```

0 rows | 1-6 of 10 columns

4.3 Generate sf Object

After cleaning the columns, we convert the dataframe into an sf object using the latitude and longitude columns. We also rename fields and adjust the coordinate reference system to match the KMZ dataset.

▶ Code

	year <int>	number <chr>	fire_center <chr>
1	2012	C10075	Cariboo
2	2012	C10118	Cariboo
3	2012	C10247	Cariboo
4	2012	C10250	Cariboo
5	2012	C20005	Cariboo
6	2012	C20009	Cariboo
7	2012	C20016	Cariboo
8	2012	C20047	Cariboo
9	2012	C20104	Cariboo
10	2012	C20108	Cariboo

1-10 of 1,108 rows | 1-4 of 8 columns Previous **1** 2 3 4 5 6 ... 111 Next

4.4 Check Duplicate

We check for any duplicate records.

▶ Code

```
The number of duplicate rows: 0
```

4.5 Check NULL

We also examine missing values in the key columns: **year**, **number**, **size_ha** or **geometry**.

▶ Code

```
The number of rows with missing values: 0
```

In the next chapter, we will compare the scraped table data with the data extracted from the KMZ files.

5 Compare Website Table Data with KMZ File Data

5.1 ID-wise Difference

The website table includes only wildfire incidents greater than 5 hectares. We begin by comparing annual wildfire counts from the website table:

▶ Code

year	count	geometry
<int>	<int>	<s_MULTIP>
2012	187	<s_MULTIP>
2013	160	<s_MULTIP>
2014	202	<s_MULTIP>
2015	216	<s_MULTIP>
2016	127	<s_MULTIP>
2017	216	<s_MULTIP>

6 rows

Next, we examine the counts of >5 hectare wildfires from the KMZ data:

▶ Code

year	count	geometry
<dbl>	<int>	<s_MULTIP>
2012	187	<s_MULTIP>
2013	160	<s_MULTIP>
2014	197	<s_MULTIP>
2015	216	<s_MULTIP>
2016	127	<s_MULTIP>
2017	220	<s_MULTIP>

6 rows

The numbers are nearly identical, except the website table includes 5 additional records in 2024, and the KMZ files include 4 additional records in 2017. These are the specific records:

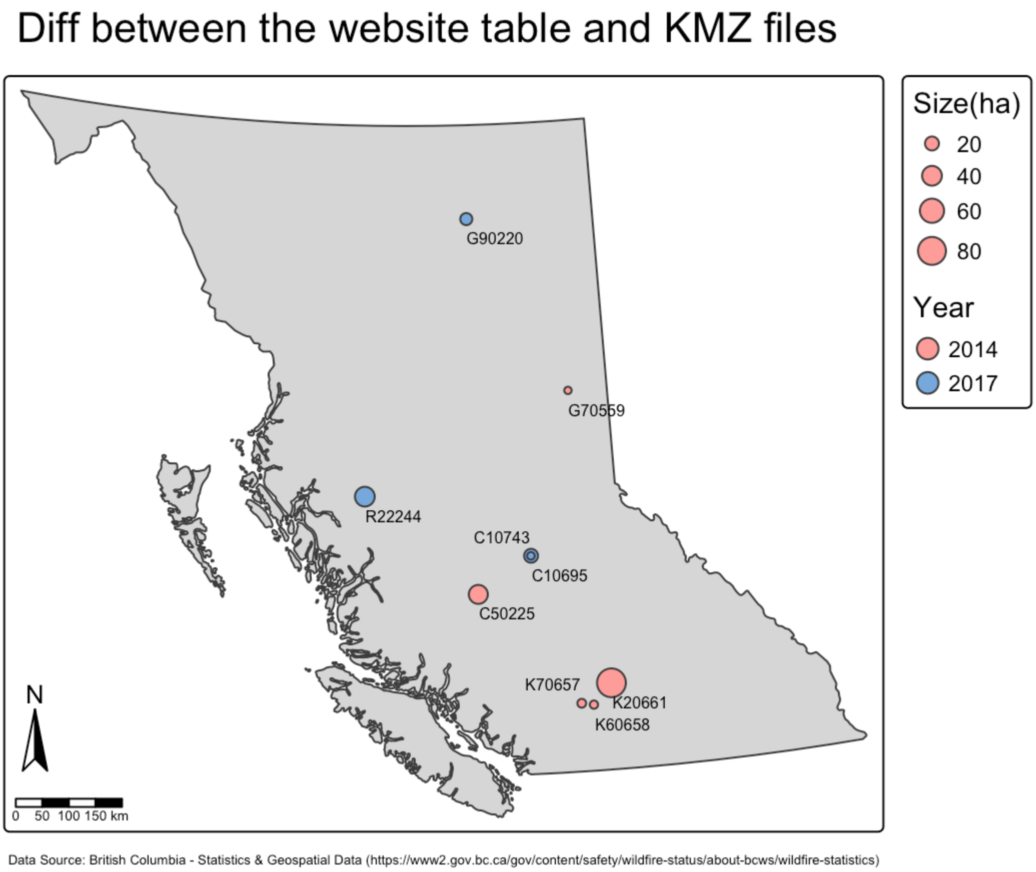
▶ Code

year	number	size_ha_tbl	size_ha_kmz
<dbl>	<chr>	<dbl>	<dbl>
2014	C50225	37.0	NA
2014	G70559	5.6	NA
2014	K20661	85.0	NA
2014	K60658	7.0	NA
2014	K70657	8.0	NA
2017	C10743	NA	6
2017	C10695	NA	20
2017	G90220	NA	15
2017	R22244	NA	40

9 rows

The geographic differences between the two datasets can be visualized on the following map:

▶ Code



We cannot clearly determine why these wildfire records appear in only one of the sources. The discrepancy may result from later data updates or inconsistencies in data management.

5.2 Size-wise Difference

There may also be records where fire sizes differ between the website table and the KMZ dataset. We examine those differences:

▶ Code

year	number	size_ha_tbl	size_ha_kmz
<dbl>	<chr>	<dbl>	<dbl>
2014	N10556	18.8	15.1

1 row

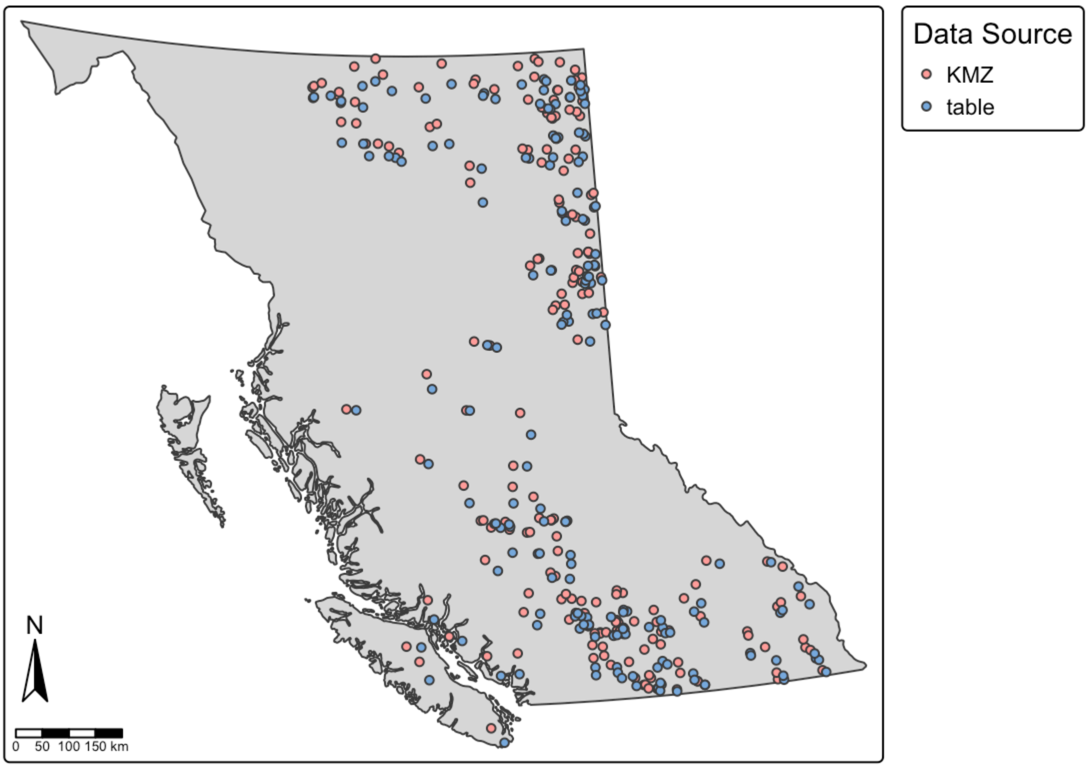
Only one record shows a size discrepancy. Again, the reason is unclear.

5.3 Location-wise Difference

While the KMZ files include geometry directly, the geometry in the website table was calculated from latitude and longitude values. This may lead to location mismatches. Below is a comparison of major wildfire locations in 2012 from both sources:

▶ Code

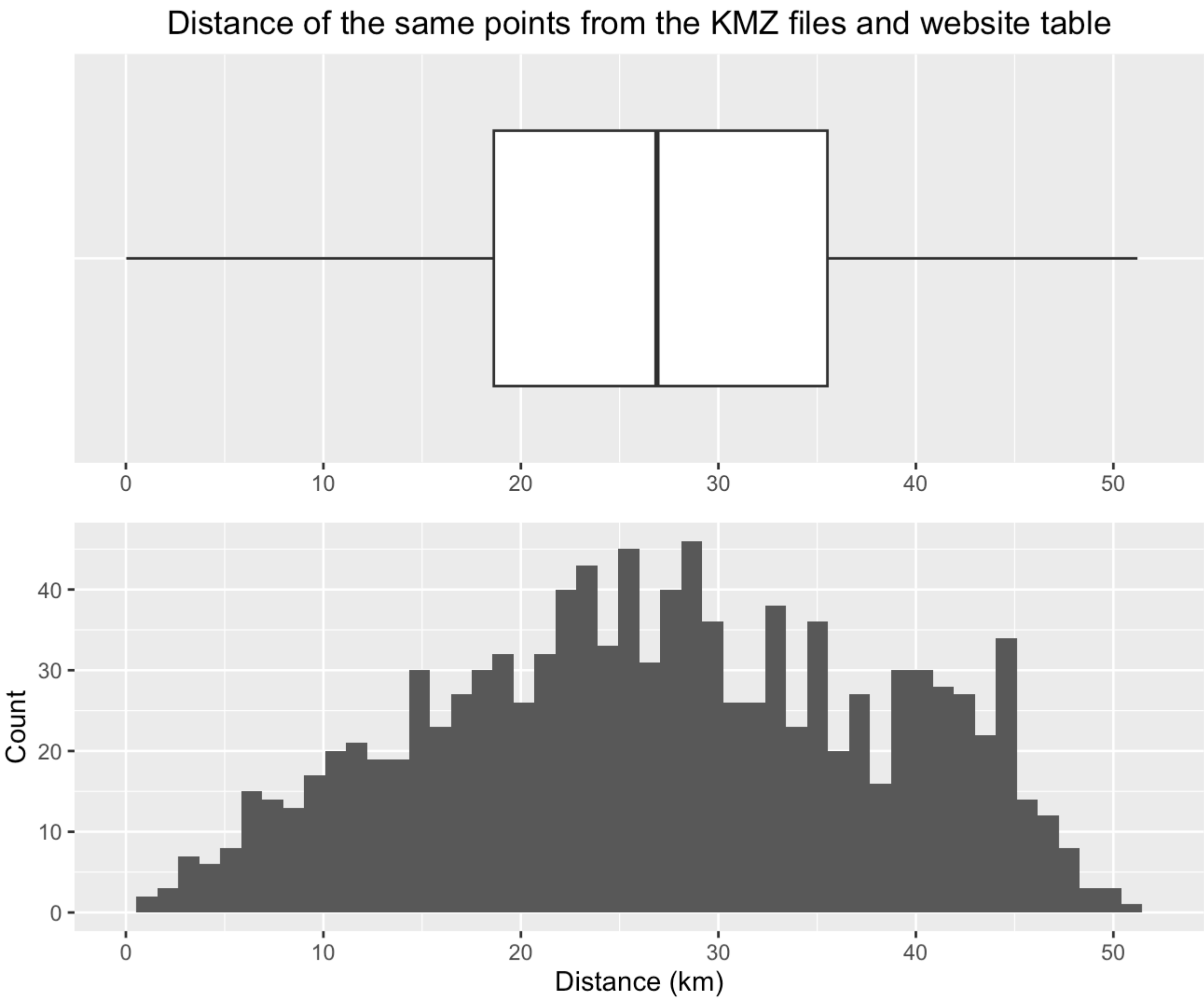
Distance of the same points from the KMZ files and website table (2012)



Data Source: British Columbia - Statistics & Geospatial Data (https://www2.gov.bc.ca/gov/content/safety/wildfire-status/about-bcws/wildfire-statistics)

As shown, corresponding points do not align spatially. The geographic differences may arise naturally when combining two different coordinate systems. To quantify the gap, we examine the distance distribution between matching records:

▶ Code



:olumbia - Statistics & Geospatial Data (https://www2.gov.bc.ca/gov/content/safety/wildfire-status/about-bcws/wildfire-statistics)

Most points differ by around 30 km, with some reaching 50 km. These differences are sizeable and suggest potential coordinate reference system issues. Due to time constraints, we cannot investigate further at this time.

6 Further Analysis

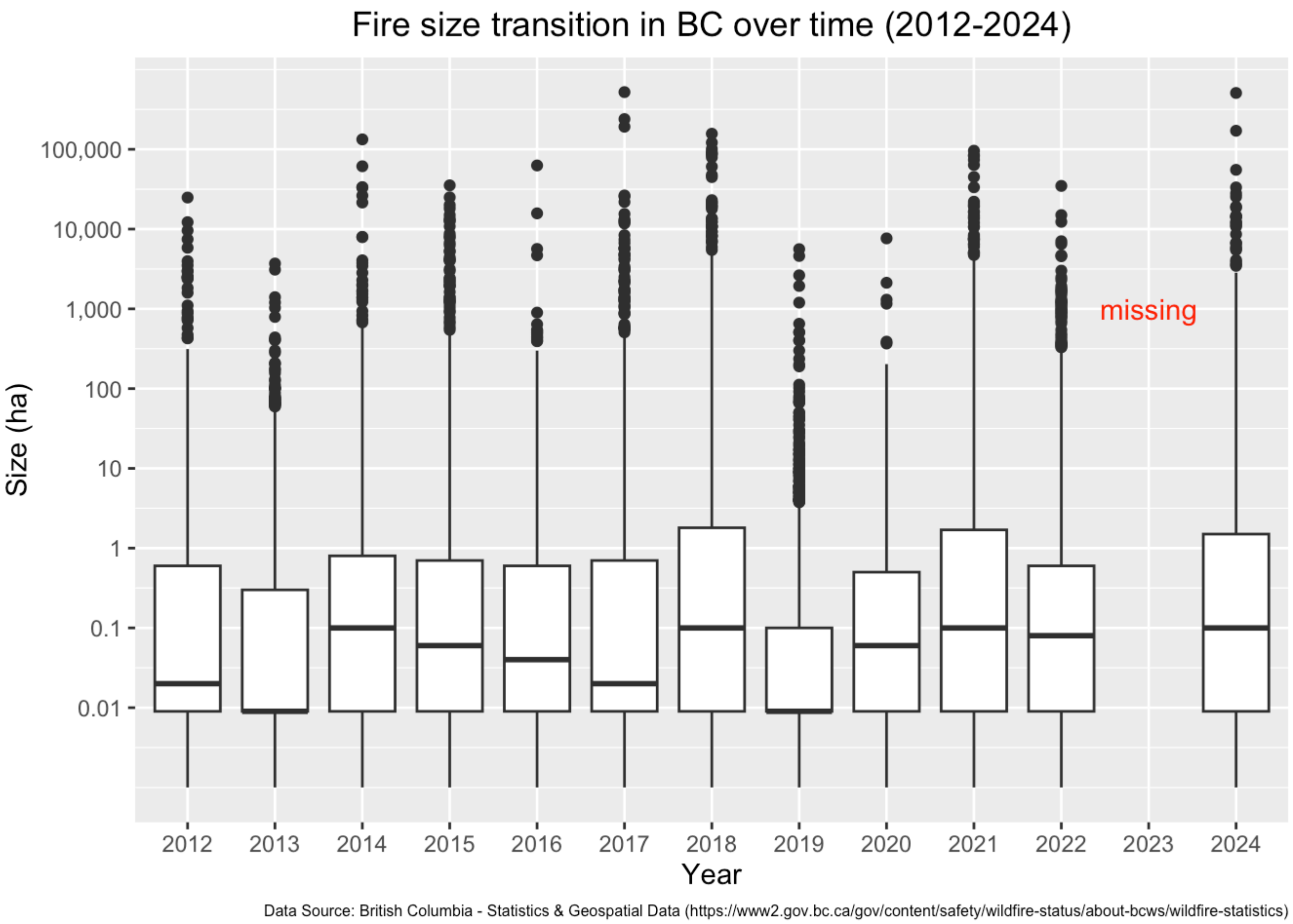
6.1 Fire Feature Transition (2012 - 2024)

We now examine changes in wildfire characteristics from 2012 to 2024 in terms of **size**, **frequency**, and **spatial distribution**. For this analysis, we use the KMZ dataset, as the scraped website table includes only 2012–2017.

6.1.1 Fire Size Transition (2012 - 2024)

First, we observe the transition of wildfire sizes over time. Although most wildfire incidents are under 1 hectare, multiple years include extremely large fires exceeding 1,000 hectares, and some even surpass 100,000 hectares. Both the median and maximum fire sizes fluctuate noticeably throughout the 13-year period. These differences by year may be influenced by weather conditions or human-related factors. However, we cannot discuss their relationship in this project because weather data is not included in our analysis.

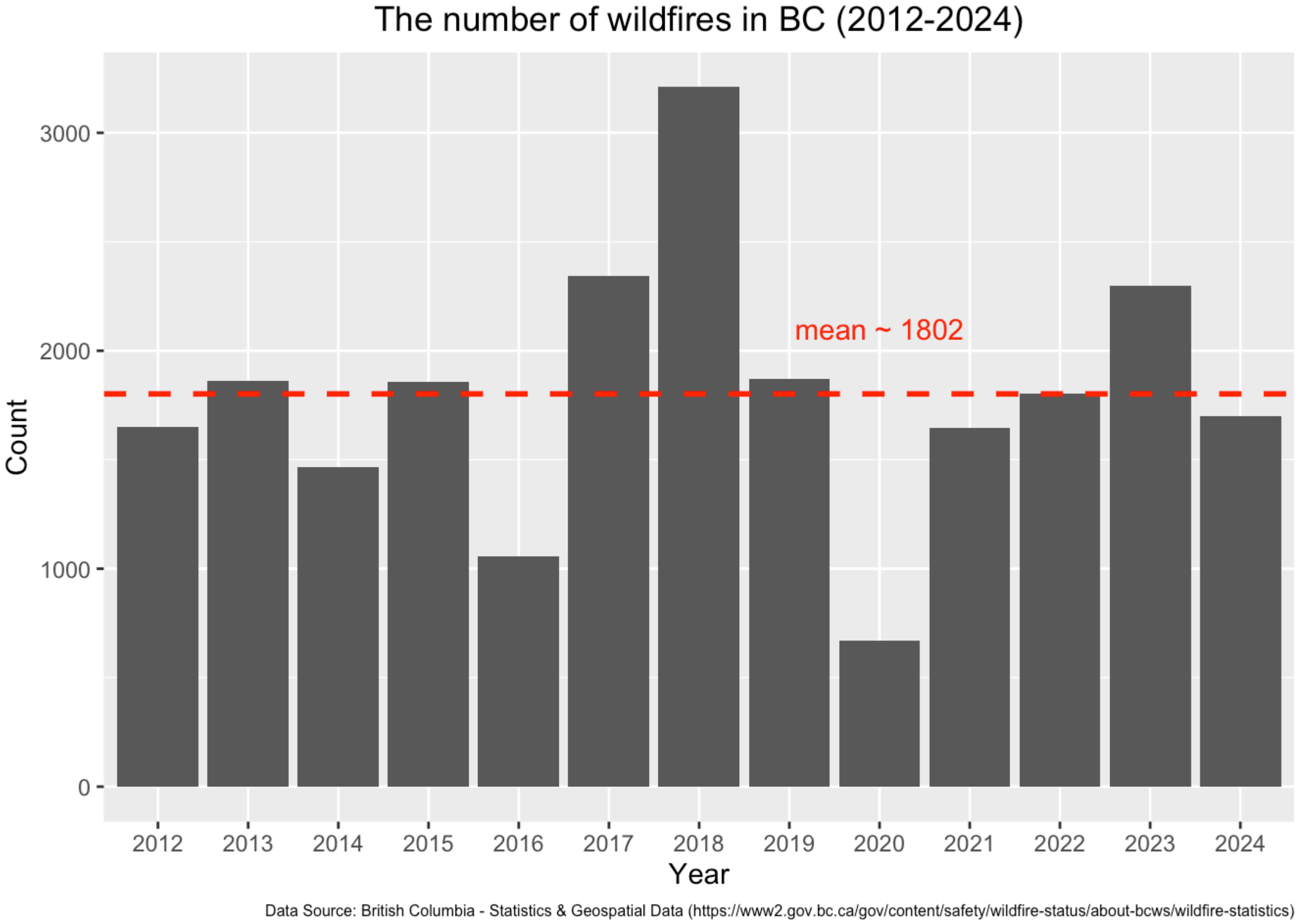
▶ Code



6.1.2 Fire Frequency Transition (2012 - 2024)

Wildfire frequency peaked in 2018, which recorded the worst year in this dataset, and reached the lowest point in 2020. Overall, the number of wildfires fluctuates around an average of approximately 1,802 incidents per year. Similar to fire size, this variation may be influenced by weather conditions or human activity, but further investigation would require supplementary datasets.

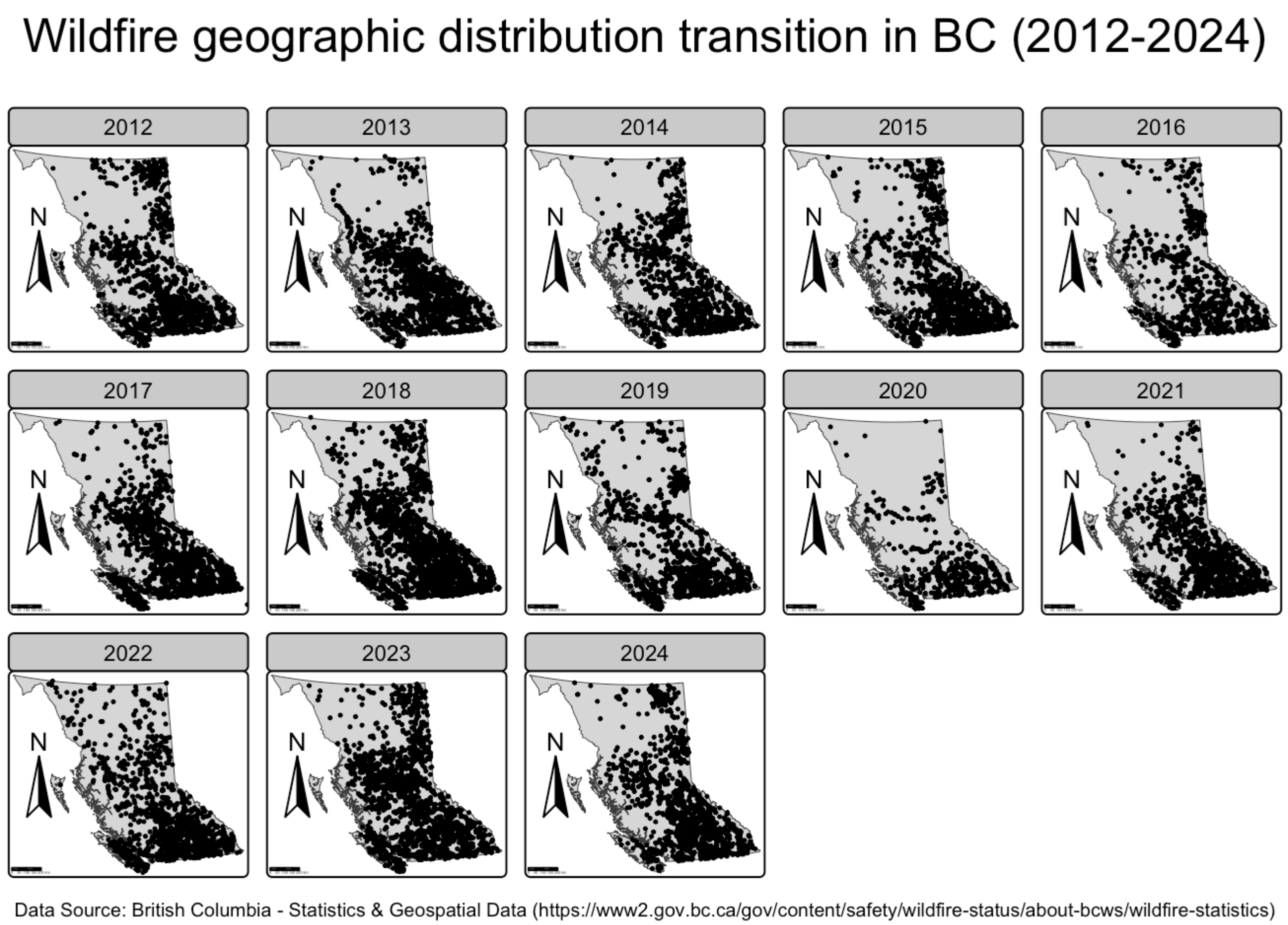
▶ Code



6.1.3 Fire Distribution Transition (2012 - 2024)

Spatial distribution patterns remain largely consistent across years. Most wildfires occur south of the central latitude of BC or in the northeastern region. This pattern persists throughout the entire observed period. To understand what drives these spatial differences, additional geographic data, such as vegetation type, population density, or infrastructure networks, would be necessary.

▶ Code



6.2 Cause

In this section, we examine how different causes relate to wildfire characteristics. Cause information is available only in the KMZ dataset and only for selected years. Below are the counts of records that include cause information by year:

▶ Code

	year	n	geometry
	<dbl>	<int>	<s_MULTIP>
1	2012	1649	<s_MULTIP>
2	2013	1861	<s_MULTIP>
3	2014	1465	<s_MULTIP>
4	2015	1858	<s_MULTIP>
5	2016	1057	<s_MULTIP>

5 rows

The recorded causes fall into three categories: **Human**, **Lightning**, and **Person**:

▶ Code

```
The unique records of cause: 'Human', 'Lightning', 'Person', 'NA'
```

Since **Human** and **Person** represent the same category, we merge them under **Human**.

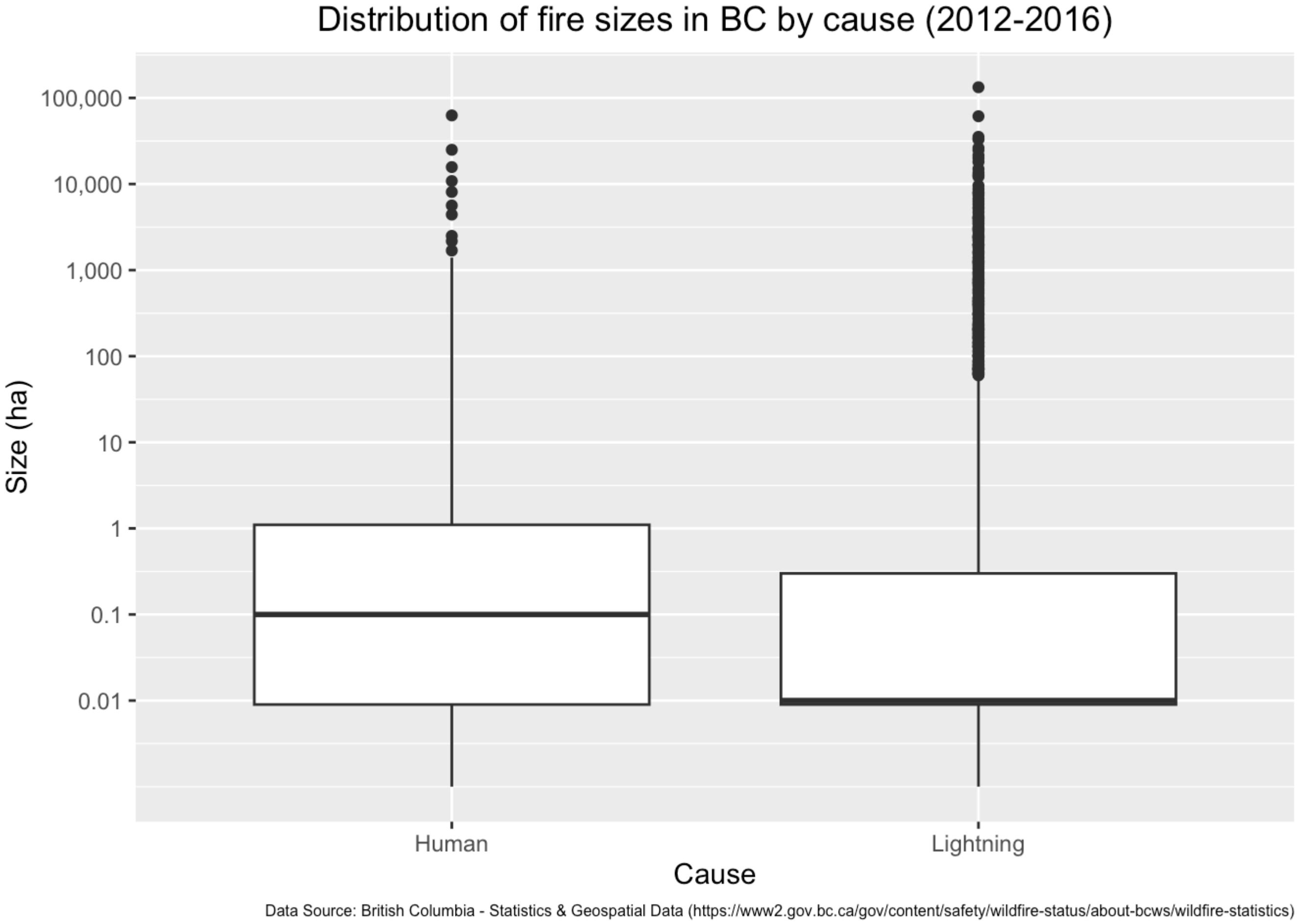
▶ Code

```
The unique records of cause: 'Human', 'Lightning', 'NA'
```

6.2.1 Cause x Fire Size

We first examine fire size distributions by cause. Both human-caused and lightning-caused wildfires can grow extremely large (over 10,000 hectares). However, lightning-caused fires more frequently remain small relative to human-caused fires:

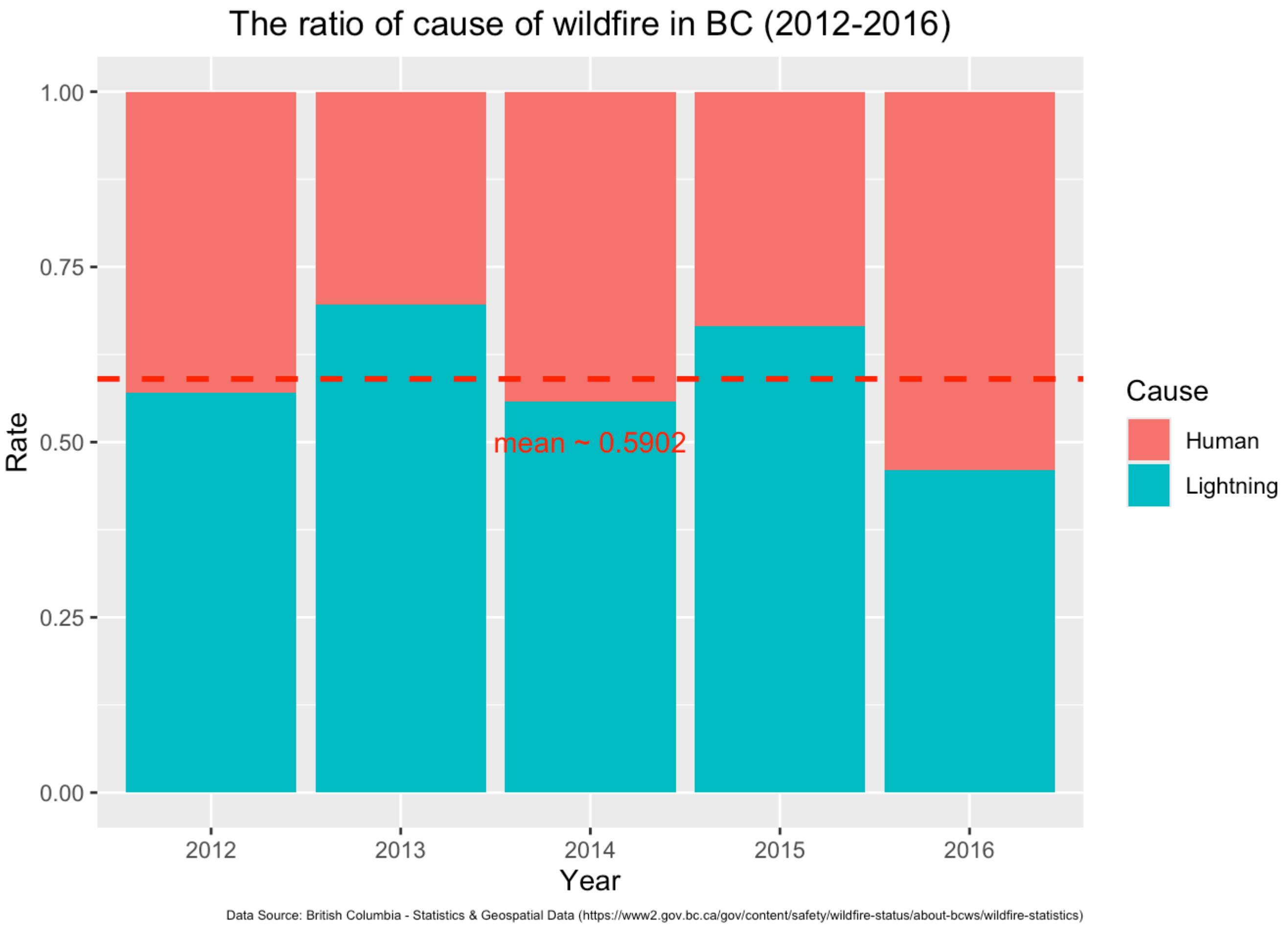
▶ Code



6.2.2 Cause x Fire Frequency

Overall, most wildfires are caused by lightning, accounting for approximately 60% of recorded incidents. Cause information is available only up to 2016, so it is difficult to evaluate whether the high fire frequency in 2018 or the low frequency in 2020 relates to differences in causes. If cause data continued beyond 2016, this comparison could clarify those trends.

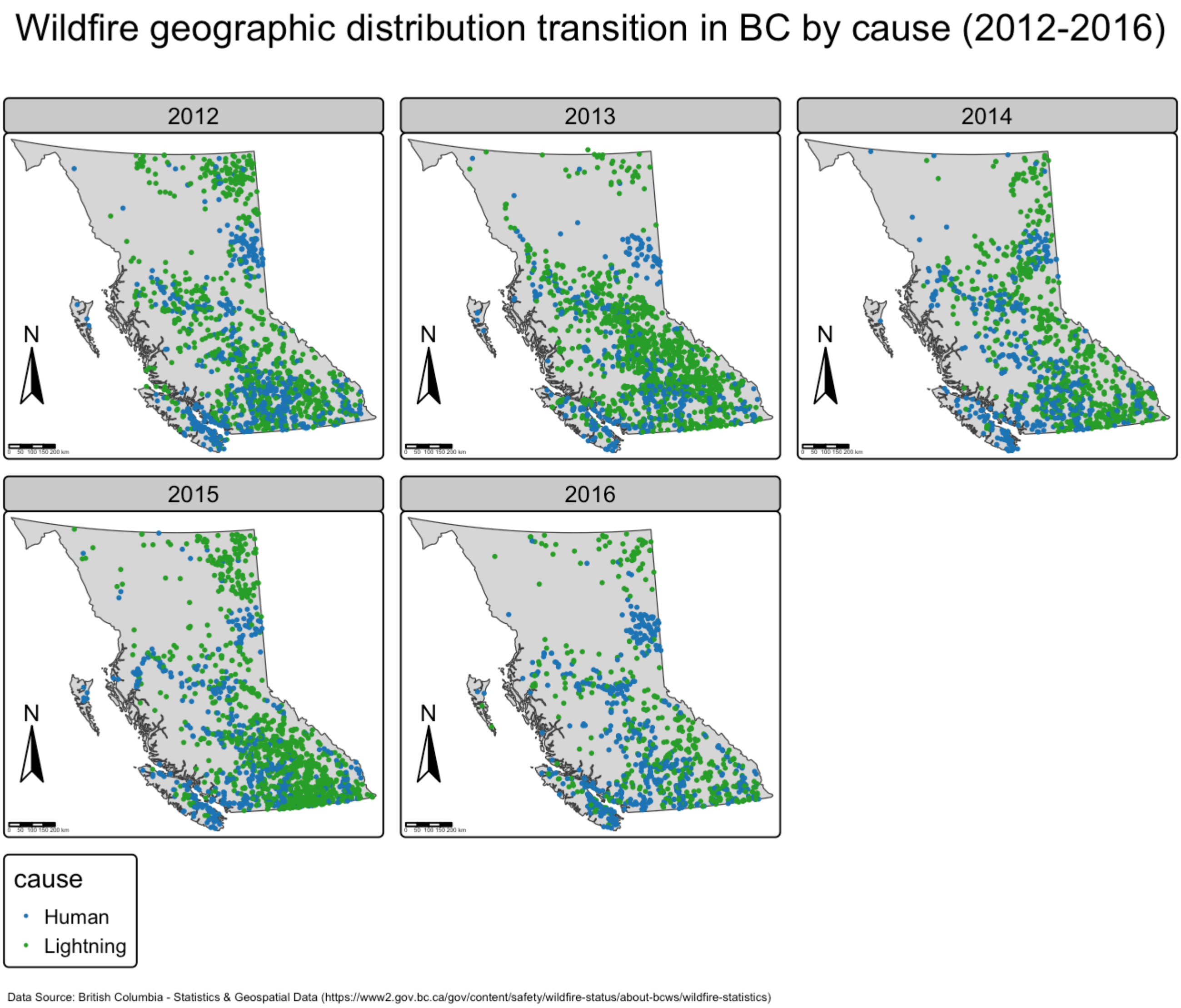
▶ Code



6.2.3 Cause x Fire Distribution

The spatial patterns of human-caused and lightning-caused wildfires appear broadly similar. Human-caused fires likely occur in or near populated and accessible regions, whereas lightning-caused fires can occur anywhere with sufficient burnable fuel, regardless of human presence. However, confirming these assumptions requires additional geographic datasets such as forest coverage, elevation, or population distribution.

▶ Code



7 Conclusion

In this project, we extracted and cleaned British Columbia wildfire data from KMZ files covering 2012–2024, handled inconsistencies in property names and missing information, and combined them into a single unified dataset. We also compared these KMZ data with wildfire records scraped from the BC Wildfire website, and while both sources generally aligned, they contained some differences in counts, size values, and geographic coordinates.

Using the cleaned dataset, we performed exploratory analysis to examine wildfire trends over time. Our findings include:

- Wildfire size varies substantially year to year, with several extreme events exceeding 100,000 hectares.
- Wildfire frequency fluctuates around an average of ~1,800 incidents annually, peaking dramatically in 2018 and reaching the lowest point in 2020.
- Spatial distribution patterns remain consistent, with most fires occurring in southern BC and parts of the northeast.
- Lightning is the leading cause of wildfires where cause data are available, responsible for around 60% of incidents, although both human- and lightning-caused fires can produce extremely large events.

Overall, this analysis highlights both the variability and the recurring patterns in wildfire characteristics across British Columbia. Future work could include incorporating additional datasets (such as weather, vegetation, or topography) to identify the reasons of fire feature changes over time, addressing coordinate system differences, and building predictive models to support wildfire management.